

Video Transcript

Video 6: Creating a Database in Python (12:24)

So, let's start with a student. So, let's have a student table, and what's a student going to look like? Well, let's say student 1 will look like this. We'll have, for example, an ID and that'll be a number. So, this is the student number one and we'll have the student name and so the name would be, for example, Alice. So, that would be a good student and we'd have S2 and S2 would be. Let's go with the name and Bob would give us somebody starting with B and finally, we might have somebody, Student 3, name Charles. Okay, so we have three students and now we need to understand. Okay, are we going to have a student table of some kind? So, for example, we could have students would be maybe a data structure of some kind. What should we choose?

Well, we'd probably choose a dictionary, so we'd have students, and now we've got some choices about how to represent the students. But one choice would be to put them all in a list in an array S1, S2, S3. So, that would be a reasonable way of listing the students. What would we do now for the problem sets? Well, let's for the P sets we'd have problem set 1 equals and we'd follow a very similar style. We'd have an ID. Now, instead of name, we need the Max score, and the maxScore might be an integer, say 100. So, that'd be problem set one, problem set 2. We'll do three of each of these. Problem set two, problem set 3. So, here we go and just for fun, we'll have this one out of 80 and this one out of maybe 120 and now what would we have?

Well, now we'd have P sets, so this should be students P sets and we'd have the problem sets and again, we'd put them into an array, P1, P2, P3. That would be reasonable. Now, we've got grades. That's the next one we need to figure out. So, let's do that. We're going to have grades and what did we decide? Well, we're going to have the student ID. So, let's see how this grades is going to work. We'll have student ID, so say student 1 and now we'd need more, right?

We'd need also the problem set ID, so pid, so that might be one as well and finally we need the score. What did they get? And so, let's suppose they got 90 on that. So, that would be one grade, grade 2 and now we'd have our grades, for example, would look like we said we'd have grades and now we'd have an array, grade one, grade two, grade 3. Just make sure our structure is correct, right? That will work. An array, we can pick out grades.

Let's think out now about our database. Let's create a database. So, we'll have a database that looks like the following. Let's think about that. We want the tables in there. So, the tables that we want are. We want students, that's a table, that's an array rather and then we want to have psets. Let's make these strings. They're going to be keys, psets, and that would be another array of P1, P2, P3 and finally we'd have grades and that would be another array. So, that's what our database would look like. Let's print it out.

Okay, so let's have this as our database, but let's actually create it. Let's get ourselves, for example, a function that would create this database and then we can start filling in these values for the database. So, we'd have, for example, we define an empty database and we'd return and we'd return this structure. That would be a reasonable way to create our database. Now, we'd say `db = empty` and fire it and now we'd have our database, perfect.

Now, we need to add students to the database. How would we do that? Well, we need add student and we'd pass in the database and now we'd `addStudent`. So, what do we need for that? Well, we need to define a student. What else do we need to define? Well, we need the `studentId` and their name and now what would we do? Well, we'd add that to our database, so we do `db`. we need to insert it. So, we depend into the database and we append something of like the student that looked like this.

So, we're going to append. Something that looks like this. So, we put `studentId` and student name. Maybe we'll have we'll go for camel case and this would be student name. Okay, so that would add a student. Should we try it? So, add `Student` and now we need to pass the database. We've got the database, it's empty and we need to pass in the `studentId`. So, we need to pass in one and for example the name of the student which we had Alice. That would add Alice to the database. Let's just do a couple more Bob and Charles. Okay, let's print out the `db`. What do we need? How do we access that? We need this correct now. `db[ 'students' ]`. Perfect. So, we need to be aware of what exactly we're appending to.

So, that got us our students. We should probably check that these IDs are unique, but we'll do that later. So, you can see that we need to now do the same and let me, I'll do it fairly quickly. We need to add problem sets and add the grades.

Let's take a look at those. So, here I've added the pset. We passed the database, the problem set ID and the total points. So, now we append to pset ID and the points and here we do add `Pset` and that problem set one total points is 100. Problem set 290-3120. Let's do the same now for

adding the grade. So, to add the grade we'll need the DB again, the student ID, the problem set ID and the number of points, and we append to grades, and let's add some grades in.

